

Harsh Bordekar

Fatigue & Damage Tolerance Engineer (d/f/m)

PhD Candidate at DLR & TU Braunschweig, expected [x]

Fatigue & Damage Tolerance: Composite & Hybrid Structures, Metallic Structures, Model V&V

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📍 Braunschweig, Germany

🌐 [Harsh's LinkedIn](#)

🏠 Permanent Resident - Germany

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PROFILE

Structural analysis and simulation engineer with 7+ years of experience in aerospace structural integrity, spanning fatigue and damage tolerance (F&DT) of composite and hybrid structures alongside metallic structures. Experience includes analyzing and validating structures against defined F&DT methods: deriving allowable defect/damage limits for a target fatigue life using small-defect fatigue models (Murakami, El-Haddad) and certification-aligned failure criteria, coupling defect measurements with numerical simulation to establish acceptance limits for damaged or defective structures, and validating theoretical damage models against measured data. This sits alongside high-fidelity linear and non-linear FEM, progressive damage analysis of composites (CDM, PDA, CZM), model verification & validation, and stochastic/uncertainty analysis for certification-oriented sizing. Currently a PhD Candidate at DLR & TU Braunschweig, developing a certification-aligned damage/leakage failure criterion that translates multiscale damage models into structural sizing inputs consistent with regulatory boundary conditions. Regularly collaborates across design, materials, and stress interfaces: reviewing externally and student-produced structural analysis work, defining technical scope for that work, and communicating findings to design and project teams. Familiar with analytical stress methods (Niu, Bruhn) alongside FE solvers (Nastran/Patran, ABAQUS), with a strong record of process automation using Python and HPC pipelines and an active interest in applying AI to improve analysis workflows.

Core Competencies: Fatigue & Damage Tolerance (F&DT), Composite & Hybrid Structures, Damage/Defect Acceptance Limits, Fatigue Life & Allowable Defect Size (Murakami, El-Haddad), Metallic & Composite Structural Analysis, Linear & Non-linear FEM (Nastran/Patran, ABAQUS), Progressive Damage Analysis (CDM, PDA, CZM), Analytical Stress Methods (Niu, Bruhn), Verification & Validation, Certification-Oriented Sizing, Simulation & Reporting Automation, Uncertainty Quantification (UQ), Python Programming, HPC (SLURM)

PROFESSIONAL EXPERIENCE

Structural Analysis / Simulation Engineer

DLR & TU Braunschweig

Composite F&DT, damage acceptance limits & validated-method sizing

Feb 2021 - Present

- (PhD) Derived a certification-aligned leakage failure criterion (leakage envelope) that maps combined strain/load states to leakage onset, defining an allowable-damage-limit philosophy per validated methods, and translating multiscale FEM outputs into structural sizing inputs consistent with regulatory boundary conditions, providing the failure-criterion definition and stress justification required for structural approval.
Tools: ABAQUS, Python, FEM post-processing using C++
- (PhD) Developed a probabilistic, physics-based multiscale framework for leakage onset prediction in CFRP hydrogen storage tanks, coupling microstructure-informed mesoscale damage models with stochastic spatial defect distributions to propagate failure across ply, laminate, and structural scales and estimate probability of failure.
Tools: Git, Docker, Python, ABAQUS, FEM, HPC/SLURM
- Performed uncertainty quantification to generate leakage probability vs. strain relationships, validating theoretical damage models against measured/simulated data and enabling risk-informed design decisions for next-generation CFRP pressure vessels under hydrogen storage conditions.
Tools: Python (scipy, numpy), Monte Carlo methods, ABAQUS
- Verified and technically reviewed externally and student-produced analysis models, meshes, and simulation results against defined methods and acceptance criteria, providing sign-off-style checks on structural analysis data and documentation and communicating findings to design and project teams.
Tools: ABAQUS, Nastran, Python
- Developed an Agentic AI application for microstructure-informed property prediction, accepting natural language queries and microstructure images as inputs to predict homogenized/effective material properties, powered by a physics-based CNN as a custom tool and a hierarchical RAG pipeline for context-aware retrieval of materials knowledge, enabling on-demand computational homogenization without full-scale FEM runs.
Tools: Python, Physics-based CNN, Hierarchical RAG, LLM agent framework, image processing
- Evaluated thermal protection effectiveness for the Callisto reusable rocket program's metal upper housing bay, executing large-scale thermal and structural simulations on HPC infrastructure.
Tools: ABAQUS, SLURM (HPC), Python, Nastran, Patran

- Instructed 60+ Master's students in FEM theory and ABAQUS-based CFRP modelling, covering Continuum Damage Mechanics (CDM), Progressive Damage Analysis (PDA), and Cohesive Zone Modelling (CZM), demonstrating both deep technical expertise and the ability to lead and communicate complex simulation workflows.
Tools: ABAQUS, Automation using Python

Research Intern, Metallic Fatigue & Damage Tolerance

DLR-Braunschweig

Damage/defect acceptance limits & fatigue-life qualification

Nov 2019 - Jan 2021

- Coupled defect measurements with numerical simulation to establish a defect-tolerance qualification methodology, translating fracture-mechanics and fatigue criteria into allowable-defect acceptance limits for manufactured metallic parts, providing the stress-analysis basis for accepting or rejecting defective structures.

Tools: FEM (ABAQUS), fracture mechanics, Python

- Assessed fatigue and defect tolerance of aerospace-grade titanium (metallic) components, deriving allowable defect sizes for a target fatigue life using small-defect fatigue models (Murakami, El-Haddad), linking as-built defect populations to structural fatigue substantiation.

Tools: Small-defect fatigue models (Murakami, El-Haddad), Python (NumPy/SciPy)

- Developed a CT-based, explainable-AI defect-detection framework providing interpretable confidence outputs for engineering sign-off, automating classification of internal defects feeding the fatigue assessment.

Tools: Python (scikit-learn / TensorFlow, SHAP / LIME / Grad-CAM), CT image processing

Student Research Assistant

Leibniz University Hannover

Institute of Static and Dynamic

Sep 2018 - Oct 2019

- Developed an ML framework for automated detection and quantification of fiber undulations in unidirectional CFRP laminates, generating spatially resolved defect metrics as direct inputs for probabilistic structural simulation models to assess manufacturing-induced variability on mechanical performance.

Tools: Matlab, Python (scikit-learn / TensorFlow), image processing, NumPy/SciPy

Design Engineer

Gujarat Industrial Development Corporation

AK. Alloys

Mar 2017 - Mar 2018

- Designed sand cast components with manufacturing tolerances and process constraints, translating functional requirements into production-ready geometries compliant with sand casting process limitations, draft angles, parting line constraints, and dimensional quality standards.

Tools: CREO, SolidWorks, AutoCAD

- Managed witness casting and end-to-end quality assurance processes, ensuring dimensional conformance and material integrity of cast components against engineering specifications and customer acceptance criteria.

Tools: CREO, SolidWorks, AutoCAD

- Planned and optimized sand casting production schedules based on capacity, material availability, and delivery requirements, improving throughput alignment with manufacturing targets. **Tools: SolidWorks, AutoCAD, ERP / production planning tools**

SKILLS SUMMARY

Fatigue & Damage Tolerance	Composite & hybrid structures F&DT Fatigue-life assessment Allowable defect/damage size (Murakami, El-Haddad) Damage/defect acceptance limits Certification concepts
Stress Analysis & FEM	FE solvers (Nastran/Patran, ABAQUS) Analytical methods (Niu, Bruhn) Linear & non-linear FEM Verification & Validation
Numerical Simulations	Anisotropic material modeling and FEM Fracture Mechanics Progressive damage analysis Multiscale Modeling
Programming Languages	Python C++ FORTRAN MATLAB
Data Analysis & Scientific Computing	NumPy Pandas Matplotlib Seaborn Plotly SALib SciPy
ML & AI	Scikit-learn TensorFlow PyTorch
LLMs & Generative AI	Hugging Face Transformers (Fine-tuning) Prompt Engineering Embeddings RAG LangChain LangGraph MCP
APIs & Backend Development	RESTful APIs FastAPI JSON
Version Control & Reproducibility	Git Docker
Application Development	VS Code JupyterLab Google Colab Streamlit Gradio
Languages	English (C1) German (B1)

AWARDS AND EDUCATION

🏆 2nd Prize: Best Paper Award

DLR (German Aerospace Center)
2026

🏆 3rd Prize: Best Paper Award

DLR (German Aerospace Center)
2024

M.Sc. Computational Methods in Engineering

Leibniz University Hannover, Germany
April 2018 – Dec 2020

Focus: Numerical Methods, Scientific Machine Learning, Deep Learning

SELECTED PUBLICATIONS

1. **Bordekar, H.**, et al. “Microstructure-informed Probabilistic Multiscale Modelling of CFRP via Machine-Learning-Derived Stochastic Volume Elements.” *Manuscript in preparation*, 2026.
2. **Bordekar, H.**, et al. “MicroStructAgent: A Physics-Informed Agentic AI Framework for Computational Homogenization and Quality Assurance of Fiber Composite Laminates.” *Manuscript in preparation*, 2026.
3. **Bordekar, H.**, et al. “Microstructure Characterization of Fiber Composite Laminate Using eXplainable Artificial Intelligence.” *Journal of Composite Materials*, vol. 60, no. 1, 2026, pp. 3–25. [\[Link\]](#)
4. **Bordekar, H.**, et al. “eXplainable Artificial Intelligence for Automatic Defect Detection in Additively Manufactured Parts Using CT Scan Analysis.” *Journal of Intelligent Manufacturing*, vol. 36, no. 2, 2025, pp. 957–974. [\[Link\]](#)

REFERENCES

Prof. Dr. Christian Huehne

Abteilungsleiter Funktionsleichtbau. Deutsches Zentrum für Luft- und Raumfahrt. Institut für Systemleichtbau

Prof. Dr. Oliver Voelkerink

Head of Institute., IMA, TU-Braunschweig

Dr. Nicola Cersullo

R&T Project Manager @ Airbus Hamburg